

COMPUTER VISION · CO1 · TECHNICAL PRESENTATION

CCD vs CMOS Sensors

Engineering Trade-offs in Image Sensing Technology

Course Code: DSA0204
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Why the Sensor Choice Matters

1

Image Quality

Noise floor and dynamic range set the ceiling on what any downstream CV algorithm can recover from a frame.

2

Power Budget

Battery-powered and embedded vision systems cannot absorb high per-pixel power draw.

3

Cost at Scale

Consumer devices ship millions of units — a few cents per sensor changes the bill of materials materially.

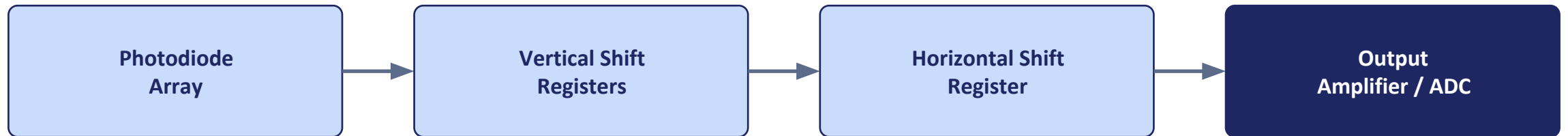
4

Speed Requirements

Real-time tracking, robotics and autonomous systems need fast, low-latency readout.

Engineering context: the choice between CCD and CMOS is rarely about which sensor is “better” — it is about which trade-offs a given application can afford.

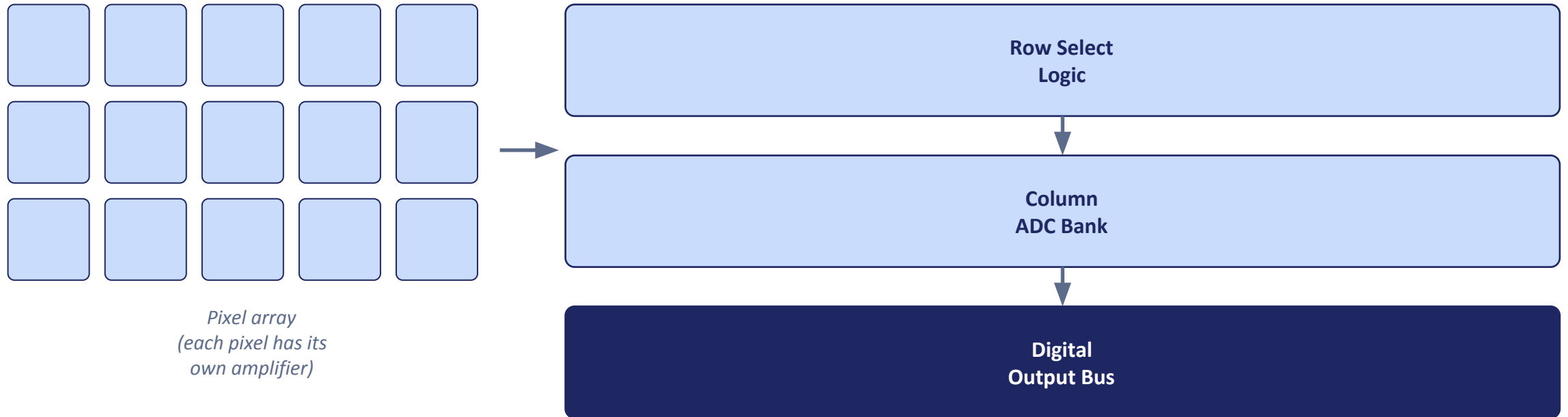
How a CCD Sensor Works



Charge is transferred physically, pixel-to-pixel, across the chip to a single output node — a serial, assembly-line process.

- Single amplifier → highly uniform pixel response, low fixed-pattern noise
- Charge-transfer readout is inherently sequential → slower frame rates
- Needs specialised manufacturing → higher cost, higher power draw

How a CMOS Sensor Works

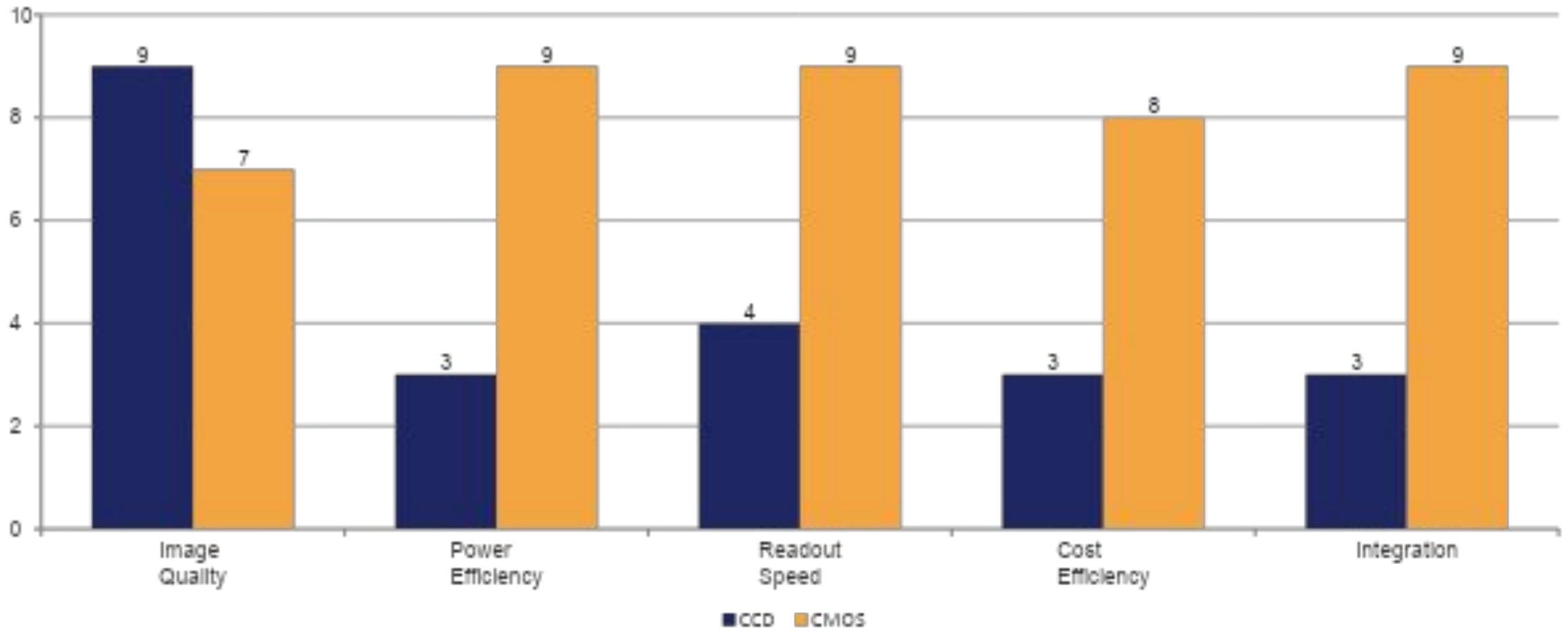


- Parallel, on-chip conversion → much faster readout and lower power
- Per-pixel amplifiers → more fixed-pattern and transistor noise
- Uses standard chip fabrication → cheaper, easier to integrate on-chip logic

Technical Comparison: CCD vs CMOS

Parameter	CCD	CMOS
Readout architecture	Serial charge transfer	Parallel, per-pixel
Power consumption	High (10–100× more)	Low
Readout speed	Slower	Fast — supports high frame rates
Noise / uniformity	Very low, highly uniform	Higher fixed-pattern noise (improving)
Dynamic range	Generally higher	Good, sensor-dependent
Manufacturing cost	Higher, specialised process	Lower, standard CMOS fabs
On-chip integration	Limited	High — logic, ADC on same die
Typical use	Scientific / astronomy imaging	Smartphones, machine vision, consumer

Scoring the Trade-offs (1-10 scale)



Matching Sensor to Application

CCD

- Astronomical imaging & telescopes
- Scientific / medical microscopy
- High-end broadcast cameras
- Applications needing extreme uniformity

CMOS

- Smartphone & consumer cameras
- Real-time machine vision / robotics
- Automotive & drone vision systems
- Battery-powered embedded devices

Conclusion

- 1 CCD and CMOS solve the same problem — converting light to signal — with fundamentally different readout architectures.
- 2 CCD favours uniformity and image quality; CMOS favours speed, power efficiency and cost at scale.
- 3 The “right” sensor is an engineering trade-off decision, driven by the target application's constraints.
- 4 Modern CMOS has closed much of the quality gap, making it the dominant choice across most Computer Vision systems today.